

Heart Disease Risk Prediction with Explainable AI: Counterfactual-Based Actionable Health Recommendations

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1 Introduction

Cardiovascular diseases (CVDs) are a leading cause of death globally, claiming approximately 17.9 million lives each year [1]. A significant percentage of these deaths are preventable through early identification of risk factors such as high cholesterol, smoking, and physical inactivity.

Machine learning models have shown strong performance in predicting CVD risk from patient health metrics, but those predictions alone are not enough for actual clinical recommendations. Traditional explainable AI methods such as SHAP [2] and LIME [3] explain why a prediction was made by attributing feature importances, rather than detailing actionable steps. Counterfactual explanations [4] address this by finding the smallest change to patient metrics that would flip the prediction from high-risk to low-risk.

In this work, we train and compare five ML models (Logistic Regression, SVM, KNN, Random Forest, XGBoost) with F_2 -score optimization. Then we design a domain-constrained counterfactual generator that respects domain knowledge such as immutable features, directional constraints, and human effort costs. Finally we deploy the pipeline as a Streamlit application with risk prediction and visual action plans. The source code for this project is available at https://github.com/lducc/AIL303m_MiniCapstone.

2 Related Works

Predicting heart disease risk is a studied problem in healthcare machine learning. Numerous models have been applied to this domain, ranging from interpretable statistical methods like Logistic Regression [5] to complex non-linear algorithms such as K-Nearest Neighbors [6], Support Vector Machines [7], Random Forests [8], and XGBoost [9]. While these models achieve high predictive performance, their lack of inherent explainability poses a significant challenge for clinical adoption.

To address the black-box nature of complex models, explanation methods like SHAP [2] and LIME [3] provide local feature attributions to interpret predictions. However, these methods only explain the model’s reasoning and do not offer actionable advice for patients. To bridge this gap, counterfactual explanations [4, 10] have shown to be a powerful tool to suggest practical changes that alter the model’s prediction. Recently, similar counterfactual explanation frameworks have been successfully applied in cardiology, such as the Generative Counterfactual ECG XAI (GCX) framework [11], which explores morphological heart changes to explain AI predictions. Our work builds upon these foundations by integrating a domain-constrained counterfactual generator into the structured predictive pipeline to provide actionable and realistic health recommendations.

3 Dataset and Feature Engineering

3.1 Dataset Overview

We use a synthetic Kaggle healthcare dataset (<https://www.kaggle.com/datasets/bertnardomariousskono/cardiovascular-disease-risk-prediction-dataset>) of $N = 15,000$ patient records with a binary target variable $\text{Heart_Disease_Risk} \in \{0, 1\}$. This raw dataset contains 16 features organized into four categories (Table 1).

Table 1: Raw feature descriptions by clinical category.

Category	Feature	Type
Demographics	Age	Continuous
	Gender	Binary
	Height_cm	Continuous
	Weight_kg	Continuous
Vitals & Labs	Systolic_BP	Continuous (mmHg)
	Diastolic_BP	Continuous (mmHg)
	Cholesterol_Total	Continuous (mg/dL)
	Cholesterol_HDL	Continuous (mg/dL)
	Cholesterol_LDL	Continuous (mg/dL)
	Fasting_Blood_Sugar	Continuous (mg/dL)
Lifestyle	Smoking_Status	Binary
	Alcohol_Consumption	Ordinal (0-2)
	Physical_Activity_Level	Ordinal (0-3)
	Stress_Level	Ordinal (1-10)
	Sleep_Hours	Discrete (4-10)
Genetics	Family_History	Binary

3.2 Exploratory Data Analysis

Prior to modeling, we performed EDA to understand feature distributions and their relationship with the target. As shown in Figure 1, the target exhibits moderate class imbalance (8,460 low-risk vs. 6,540 high-risk), motivating class-weighted training.

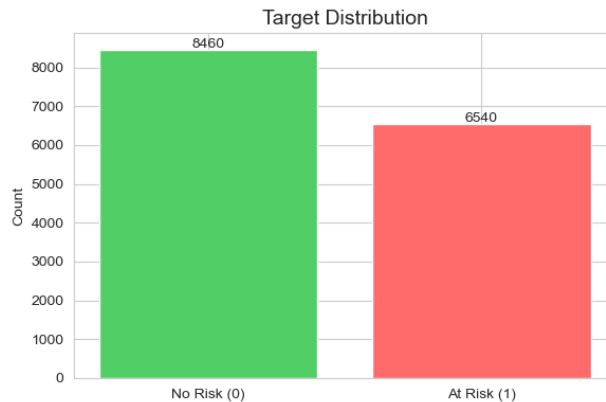


Figure 1: Target class distribution between low-risk (0) and high-risk (1) patients.

Pearson correlation analysis (Figure 2) reveals strong inter-correlation among blood pressure features (Systolic_BP and Diastolic_BP, $r = 0.81$; Systolic_BP and MAP, $r = 0.97$) and cholesterol subtypes (Choles-

terol_Total and Cholesterol_LDL, $r = 0.83$). These multicollinear groups informed the design of derived features. The target correlates most strongly with Systolic_BP ($r = 0.35$), Family_History ($r = 0.20$), Age ($r = 0.17$), and MAP ($r = 0.18$), while lifestyle features (Smoking_Status, Physical_Activity_Level) show weaker but meaningful associations.

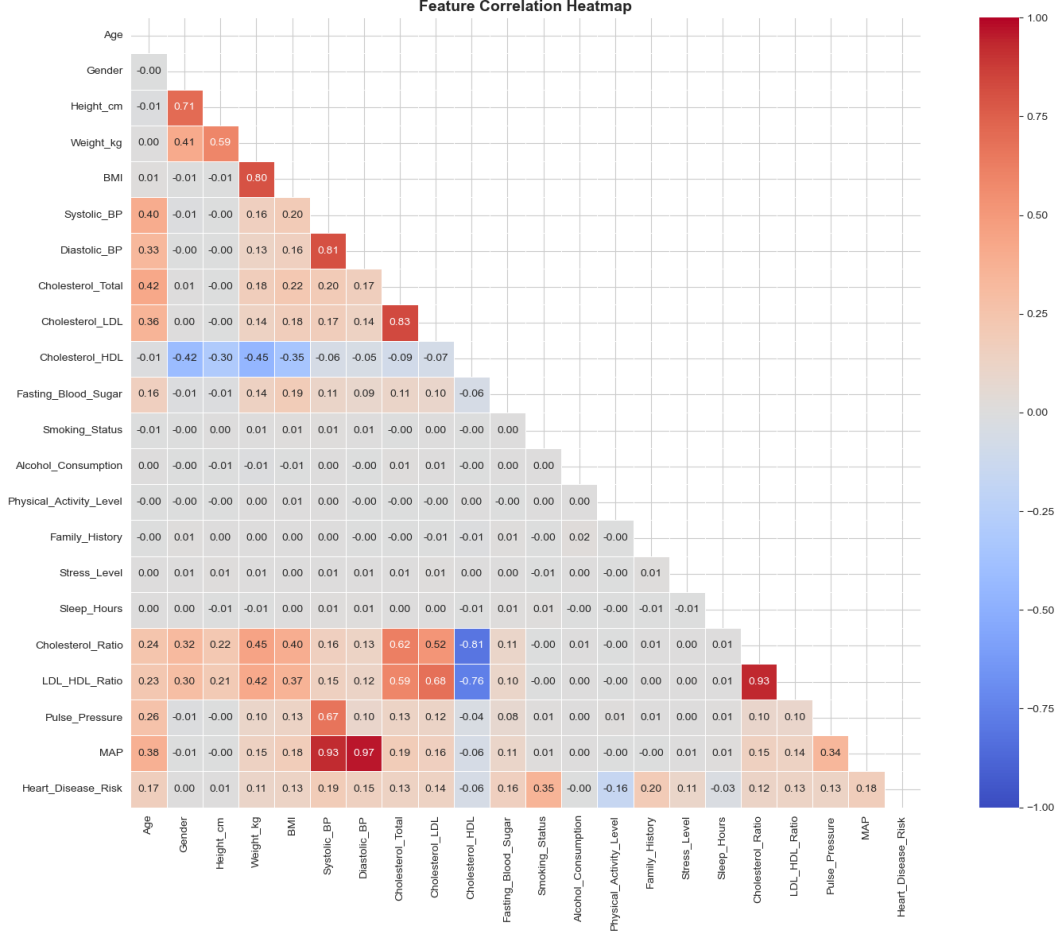


Figure 2: Pearson correlation heatmap for all features

3.3 Derived Features

We then engineer 5 additional features from the raw inputs using actual medical formulas:

$$\text{BMI} = \frac{\text{Weight_kg}}{(\text{Height_cm}/100)^2} \quad (1)$$

$$\text{Cholesterol_Ratio} = \frac{\text{Cholesterol_Total}}{\text{Cholesterol_HDL}} \quad (2)$$

$$\text{LDL_HDL_Ratio} = \frac{\text{Cholesterol_LDL}}{\text{Cholesterol_HDL}} \quad (3)$$

$$\text{Pulse_Pressure} = \text{Systolic_BP} - \text{Diastolic_BP} \quad (4)$$

$$\text{MAP} = \text{Diastolic_BP} + \frac{\text{Pulse_Pressure}}{3} \quad (5)$$

3.4 Preprocessing

We standardized the features using z-score normalization: $x_i^{\text{scaled}} = (x_i - \mu_i)/\sigma_i$, where μ_i and σ_i are computed on the training set only to avoid data leakage. The scaler is reused at inference time.

4 Methodology

4.1 Training Pipeline

All five models use the following pipeline: First they will go through splitting by stratified 80/20 train-test split, then the StandardScaler preprocessing, and GridSearchCV with StratifiedKFold ($k=5$) for hyperparameter optimization.

In medical risk prediction, missing a high-risk patient (false negative) is far more dangerous than flagging a healthy one (false positive). We use the F_2 -score as the optimization metric for this reason. The F_β -score weights recall β^2 times more than precision:

$$F_\beta = (1 + \beta^2) \cdot \frac{\text{Precision} \cdot \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}} \quad (6)$$

With $\beta=2$, recall is weighted 4 times higher than precision, prioritizing detection of at-risk patients. To address class imbalance, we use balanced class weights ($w_c = N/2N_c$) for LR, SVM, and RF, and equivalent `scale_pos_weight` for XGBoost.

4.2 Model Architectures

4.2.1 Logistic Regression

Logistic Regression [5] models the positive class probability using the Sigmoid function:

$$P(y=1 | \mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{x} + b)}} \quad (7)$$

where $\mathbf{w} \in \mathbb{R}^d$ is the coefficient vector and b is the intercept. Training minimizes the L_2 -regularized cross-entropy loss with inverse regularization strength $C \in \{0.01, 0.1, 1.0, 10.0\}$. A key advantage for xAI is that each w_i directly represents the change in log-odds per unit increase in feature x_i , enabling coefficient extraction for fast counterfactual inference (Section 4.3).

4.2.2 Support Vector Machine

SVM [7] with an RBF kernel finds the optimal separating hyperplane in kernel space:

$$f(\mathbf{x}) = \sum_{i=1}^{N_{\text{SV}}} \alpha_i y_i \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}\|^2) + b \quad (8)$$

where α_i are learned support vector weights and γ controls kernel bandwidth. We search $C \in \{0.1, 1.0, 10.0\}$ and $\gamma \in \{\text{scale}, \text{auto}\}$, with Platt scaling for probability calibration.

4.2.3 K-Nearest Neighbors

KNN [6] classifies by majority vote among the k nearest neighbors using the Minkowski distance:

$$d_p(\mathbf{x}_i, \mathbf{x}_j) = \left(\sum_{l=1}^d |x_{il} - x_{jl}|^p \right)^{1/p} \quad (9)$$

We search $k \in \{3, 5, 7, 9, 11\}$, weights $\in \{\text{uniform}, \text{distance}\}$, and $p \in \{1, 2\}$.

4.2.4 Random Forest

Random Forest [8] aggregates T decision trees, each trained on a bootstrap sample with random feature subsets. The final prediction is the majority vote $\hat{y} = \text{mode}\{h_t(\mathbf{x})\}_{t=1}^T$, where each tree splits on the feature minimizing Gini impurity $G = 1 - \sum_c p_c^2$. We search $T \in \{50, 100, 200\}$, max depth $\in \{5, 10, \text{None}\}$, and min samples per leaf $\in \{1, 2, 4\}$.

4.2.5 XGBoost

XGBoost [9] builds gradient-boosted trees sequentially, where each new tree f_t corrects the residuals of the previous ensemble: $\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta f_t(\mathbf{x}_i)$. The objective balances loss with a complexity regularizer:

$$\mathcal{L}^{(t)} = \sum_{i=1}^N \ell(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)) + \gamma T_{\text{leaves}} + \frac{\lambda}{2} \sum_j w_j^2 \quad (10)$$

We search $T \in \{50, 100, 200\}$, max depth $\in \{3, 5, 7\}$, and learning rate $\eta \in \{0.01, 0.1, 0.2\}$.

4.3 Counterfactual Explanation System

4.3.1 Problem Formulation and Domain Constraints

Given a classifier f and a patient $\mathbf{x}_0$ with $f(\mathbf{x}_0) = 1$ (high-risk), we seek a counterfactual $\mathbf{x}'$ such that: (i) $f(\mathbf{x}') = 0$, (ii) the cost-weighted distance is minimized, (iii) derived features are physically consistent, and (iv) domain constraints are satisfied.

Immutable features $\mathcal{I} = \{\text{Age, Gender, Height, Family_History}\}$ cannot be changed. **Derived features** $\mathcal{D}$ (BMI, cholesterol ratios, pulse pressure, MAP) are auto-recomputed from source features and never directly optimized. The 12 remaining **source features** $\mathcal{S} = \mathcal{F} \setminus (\mathcal{I} \cup \mathcal{D})$ are the core optimization variables.

To mimic real clinical progression, each source feature has a strict **directional constraint**: blood pressure, cholesterol, blood sugar, smoking, alcohol, and stress can only decrease; physical activity, sleep, and HDL can only increase. Furthermore, each feature is assigned an asymmetric **cost weight** c_i reflecting human effort. Behavioral changes have low effort (e.g., $c_{\text{alcohol}} = 1.0$), while metabolic changes requiring structured medication have high effort (e.g., $c_{\text{cholesterol}} = 5.0$).

4.3.2 Optimization Framework and Objective Function

Existing counterfactual generation techniques rely on generic optimization algorithms with shortcomings in medical contexts. For example, basic gradient descent methods [4] struggle with strict domain boundaries and directional constraints, frequently converging to adversarial near-boundary local minima. Mixed-Integer Linear Programming (MILP) [12] handles discrete configurations natively but becomes computationally expensive (Σ_2^p -complete generalizations) for complex ensembles and strictly assumes feature independence. Genetic Algorithms (GA) [13] are flexible but they are expensive due to population sizing and heuristic permutations, offering no global optimality guarantees. Finally, frameworks like DiCE [10] prioritize diverse options but often produce physically impossible records (e.g., reducing BMI while maintaining exact weight) and do not natively support disparate human effort costs.

To overcome these, we implement a highly specific objective function where the optimizer minimizes a delta vector $\delta \in \mathbb{R}^{12}$:

$$\min_{\delta} \underbrace{\sum_{i \in \mathcal{S}} c_i \cdot \frac{|\delta_i|}{r_i}}_{\text{Human Effort Cost}} + \underbrace{100 \cdot \max(0, P(y=1 \mid \mathbf{x}_0 + \delta) - \tau)}_{\text{Prediction Penalty}} \quad (11)$$

subject to explicit box constraints $\ell_i \leq \delta_i \leq u_i$ governing directional movement and data boundaries. The L_1 norm acts as a sparsity inducer over features, heavily scaled by the pre-calculated natural range r_i to normalize heterogeneous scales, which τ stands for the decision boundary.

We solve this using **L-BFGS-B** [14], a quasi-Newton solver natively equipped for gradient projection to satisfy hard box constraints precisely. We mitigate local optimality by employing a multi-start strategy

($R=4$), triggering one run from $\delta_0 = \mathbf{0}$ and three from localized random seed points within 20% of the constraint bounds.

Upon stabilization, lifestyle features are truncated back to discrete integers, and continuous laboratory metrics are rounded to one decimal tier. Values beneath clinically actionable thresholds (e.g., $\Delta\text{BP} < 5$ mmHg) are zeroed, ensuring purely actionable final action plans.

5 Experimental Results

5.1 Model Comparison

Table 2 summarizes the performance of all five models on the 20% holdout test set.

Table 2: Model performance on the 20% holdout test set.

Model	Accuracy	Precision	Recall	F_2	ROC-AUC
Logistic Regression	0.7283	0.6814	0.7080	0.7025	0.8040
SVM (RBF)	0.7193	0.6713	0.6980	0.6925	0.7914
KNN	0.7130	0.6959	0.6070	0.6229	0.7571
Random Forest	0.7153	0.6667	0.6942	0.6885	0.7866
XGBoost	0.7243	0.6793	0.6965	0.6930	0.7960

Logistic Regression achieves the highest F_2 -score (0.7025) and ROC-AUC (0.804), followed closely by XGBoost ($F_2 = 0.693$, ROC-AUC = 0.796). KNN underperforms on recall (0.607), making it unsuitable for a recall-sensitive medical application.

5.2 Counterfactual Explanation

We select **Logistic Regression** for deployment despite potentially higher accuracy from ensemble models, because coefficient transparency enables direct interpretation of each feature’s risk contribution.

Table 3: Example counterfactual action plan for a high-risk patient.

Feature	Original	Target	Δ	Type
Smoking Status	Smoker	Non-Smoker	−1	Lifestyle
Physical Activity	1	3	+2	Lifestyle
Stress Level	7	4	−3	Lifestyle
Sleep Hours	5	7	+2	Lifestyle
Systolic BP	160	145	−15	Medical

The optimizer prioritizes low-cost lifestyle changes before recommending medical interventions. We evaluated the DiCE framework [10] as an alternative, but found it produces physically inconsistent patients (e.g., low weight with high BMI), lacks directional constraints, and does not model human effort costs.

6 System Architecture

We deployed the model on Streamlit, which contains a model layer with five saved models (LR used at inference), and an application layer with patient input forms, Plotly risk gauge charts (before and after), a horizontal bar chart comparing current versus target values, and metric cards with delta arrows for each recommended action.

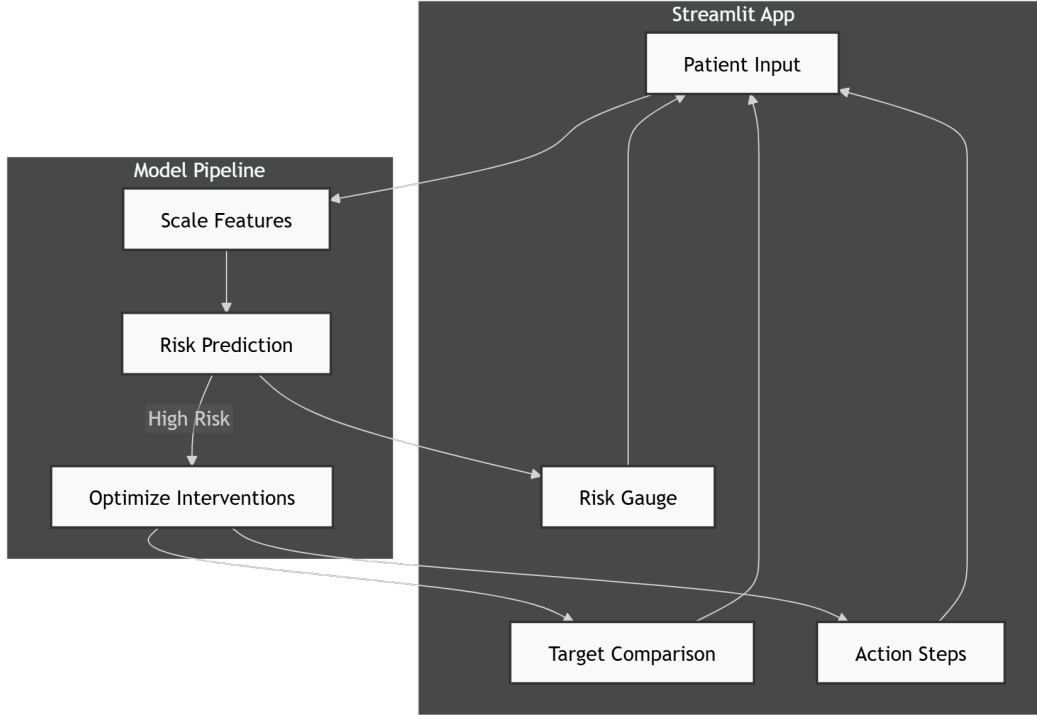


Figure 3: System model with data ingestion, counterfactual optimization, and Streamlit frontend.

7 Discussion and Conclusion

In this system we try to shift from explanatory to actionable AI in detecting heart disease. Unlike SHAP/LIME which explain why a prediction was made, our counterfactual system tells patients what actions are needed to lower down their risk value.

A major limitation and a core weakness of this project is the reliance on a synthetic generated Kaggle dataset and a relatively simple model for demonstration purposes. Because the data is artificially generated, the model’s insights might not accurately generalize to real-world clinical scenarios. Furthermore, the simplicity of Logistic Regression limits its capacity to capture complex non-linear interactions in patient health metrics. Future work must prioritize validation on real clinical datasets, such as the Framingham Heart Study, empirically learning cost weights from patient adherence data, and extending the counterfactual explainer to more sophisticated predictive models.

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